NN Introduction

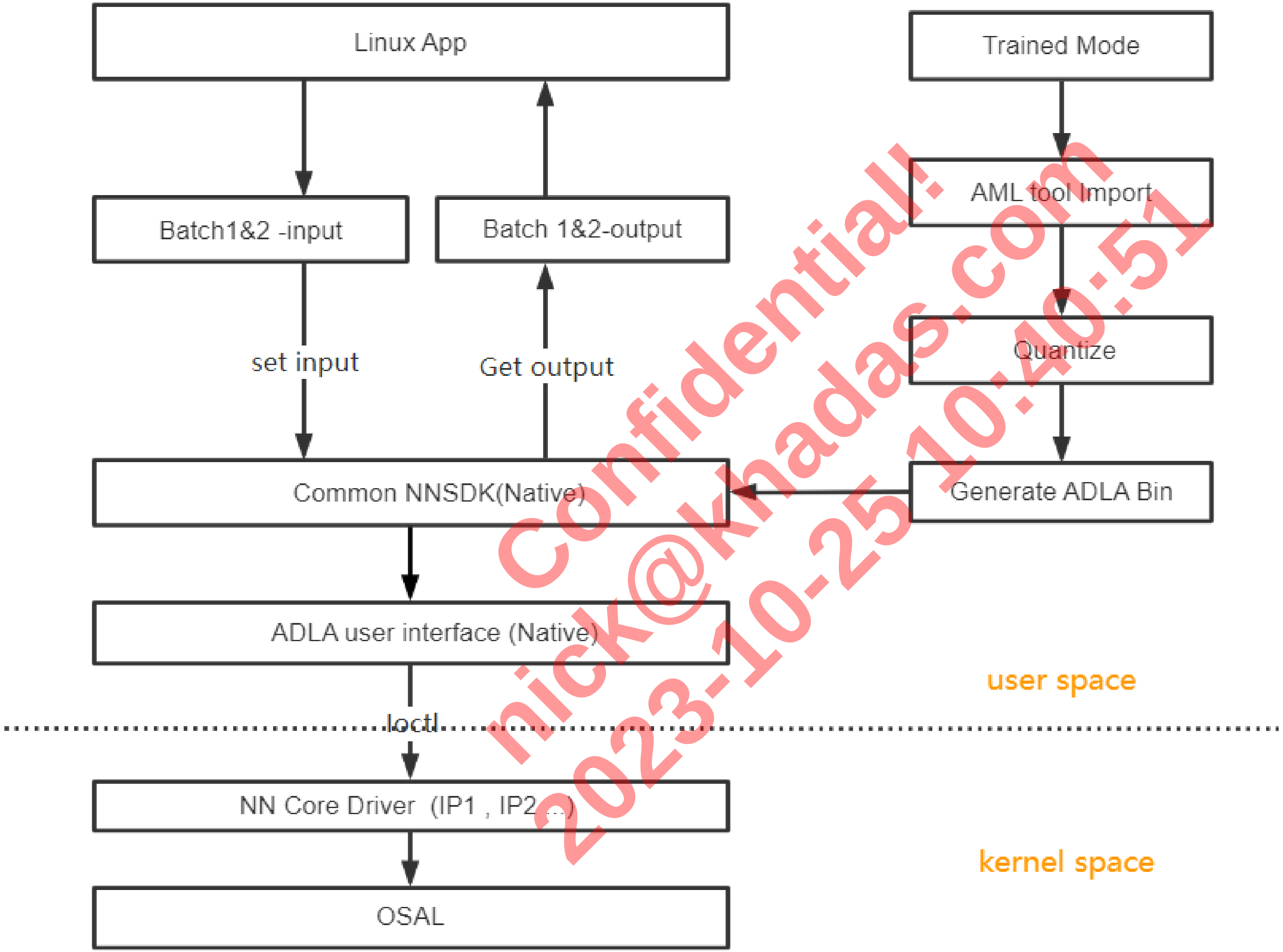
2022-10-14

Contents

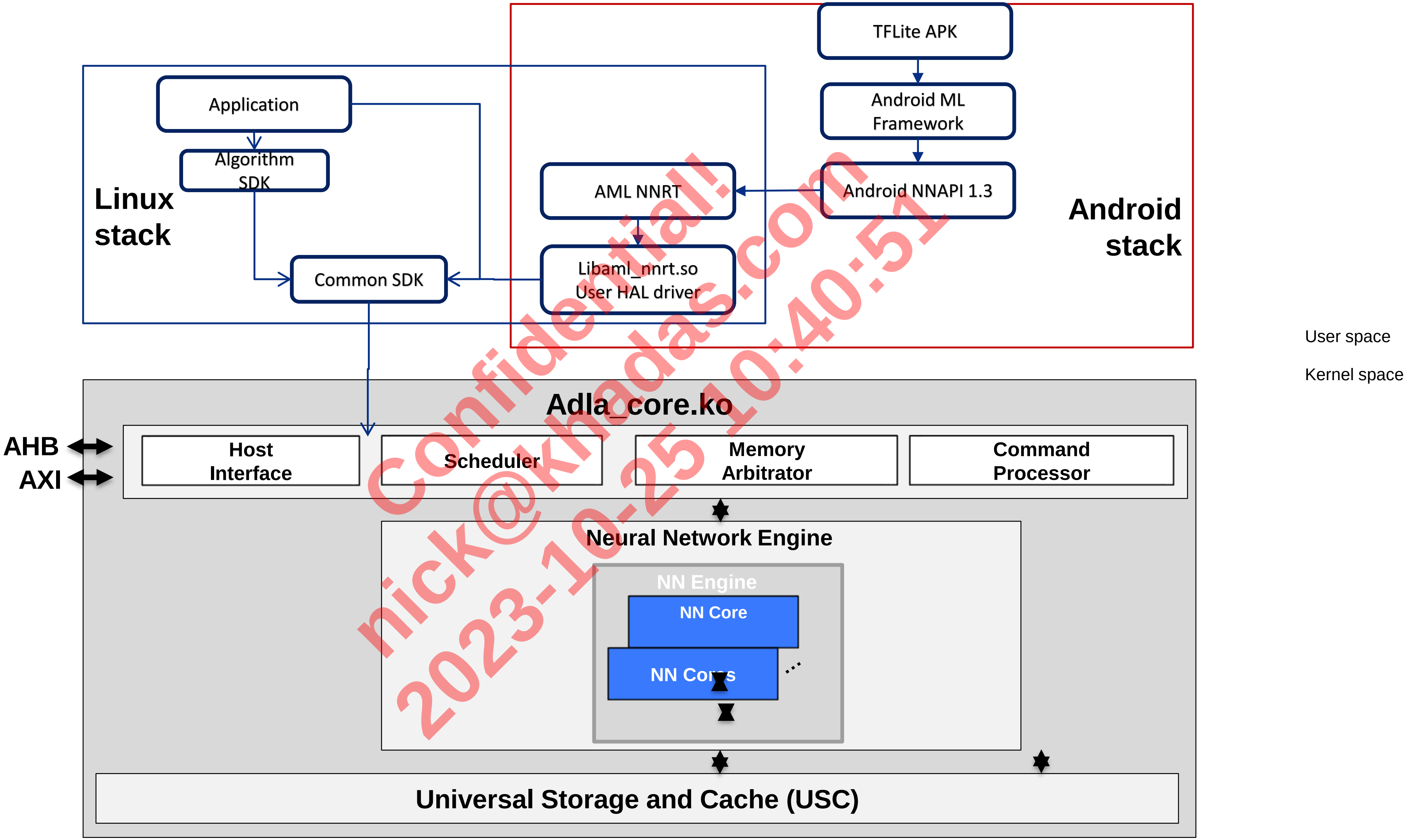
Amlogic NN Summary

- Amlogic ADLA is a neural acceleration engine. It is designed to accelerate the inference of AI image models such as image classification, detection and segmentation. It can also be used for AI voice models such as speech to text, text to speech, voice denoise, keyword detection.
- Supports caffe, tensorflow, tflite, darknet, onnx, Keras, Pytorch, Paddle, Mxnet open-source models
- Supports three quantization methods including int8, int16, uint8, etc.
- The supported open-source framework model needs to be quantified using the above quantification method first, and then convert to adla file that the chip can recognize and load
- Supports Android, Linux, FreeRTOS

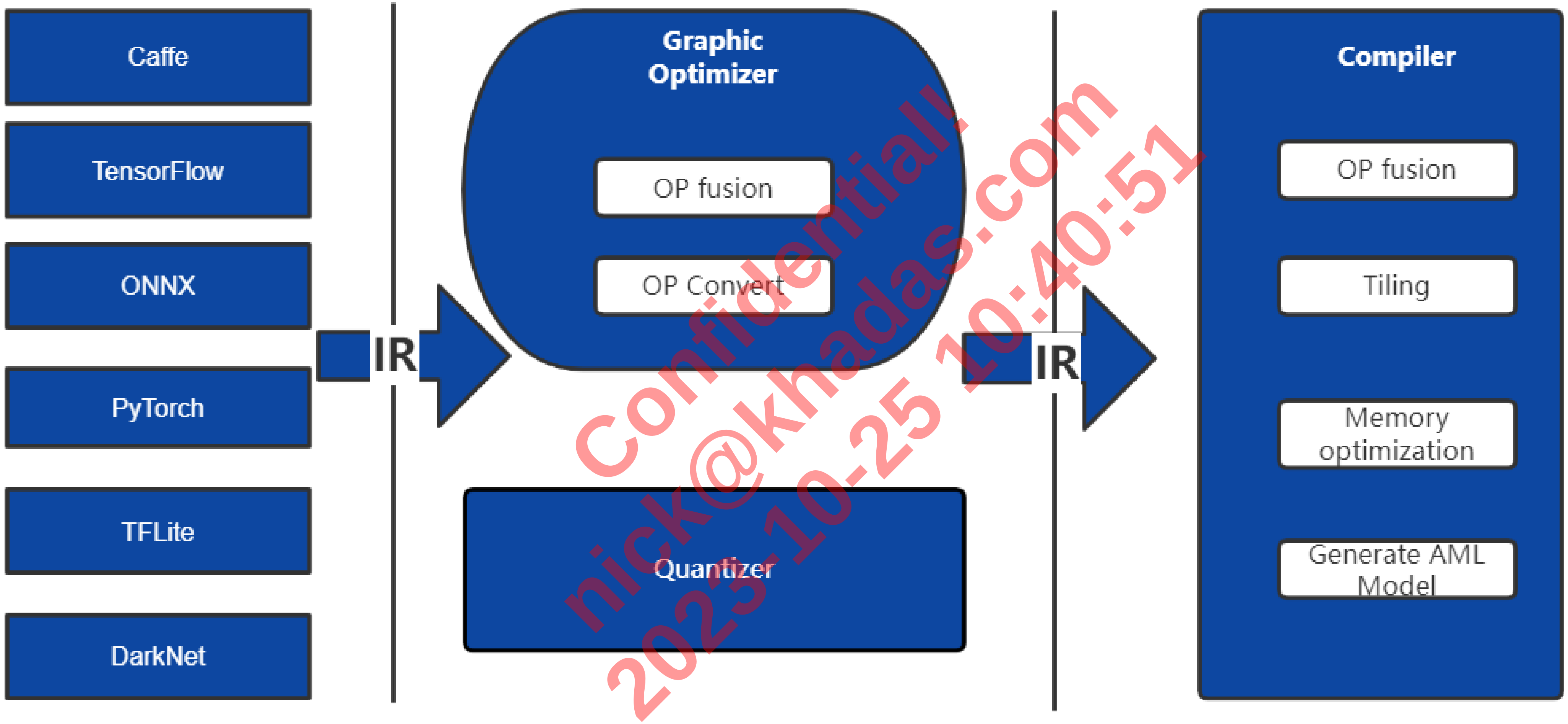
NN Software Flow



NN Software Stack



Model Convert



NN OP Support List

op name	op name	op name	op name	op name	op name	op name	op name
ABS	COS	FLOOR_DIV	LOG	NEG	REDUCE_PROD	SLICE	TANH
ADD	CUMSUM	FULLY_CONNECTED	LOG_SOFTMAX	NOT_EQUAL	RELU	SOFTMAX	TILE
ADDN	DEPTH_TO_SPACE	GATHER	LOGICAL_AND	ONE_HOT	RELU_N1_TO_1	SPACE_TO_BATCH_ND	TOPK_V2
ARGMAX	DEPTHWISE_CONV_2D	GATHER_ND	LOGICAL_NOT	PACK	RELU6	SPACE_TO_DEPTH	TRANSPOSE
ARGMIN	DEQUANTIZE	GREATER	LOGICAL_OR	PAD	RESHAPE	SPLIT	TRANSPOSE_CONV
AVERAGE_POOL_2D	DIV	GREATER_EQUAL	LOGISTIC	PADV2	RESIZE_BILINEAR	SPLIT_V	UNIDIRECTIONAL_SEQUENCE_LSTM
BATCH_TO_SPACE_ND	ELU	HARD_SWISH	LSTM	POW	RESIZE_NEAREST_NEIGHBOR	SQRT	UNPACK
BIDIRECTIONAL_SEQUENCE_LSTM	EMBEDDING_LOOKUP	L2_NORMALIZATION	MAX_POOL_2D	PRELU	REVERSE_V2	SQUARE	REDUCE_ALL
BIDIRECTIONAL_SEQUENCE_RNN	EQUAL	L2_POOL_2D	MAXIMUM	QUANTIZE	RNN	SQUARED_DIFFERENCE	
BROADCAST_TO	EXP	LEAKY_RELU	MEAN	RANGE	RSQRT	SQUEEZE	
CAST	EXPAND_DIMS	LESS	MINIMUM	REDUCE_ANY	SELECT	STRIDED_SLICE	
CONCATENATION	FILL	LESS_EQUAL	MIRROR_PAD	REDUCE_MAX	SELECT_V2	SUB	
CONV_2D	FLOOR	LOCAL_RESPONSE_NORMALIZATION	MUL	REDUCE_MIN	SIN	SUM	

NN Performance and Bandwidth

		Board: C302X			Board:T7C	
Network	Dtype	Input size	FPS	Bandwidth(MB/inference)	FPS	Bandwidth(MB/inference)
resnet50		224*224*3	102.09	30.88	168.66	33.02
mobilenet		224*224*3	386.25	3.32	451.06	4.17
inception_v1		224*224*3	212.36	9.63	312.01	10.76
inception_v2		224*224*3	178.73	13.73	248.08	15.88
inception_v3		299*299*3	64.65	29.95	126.89	35.1
inception_v4		299*299*3	32.51	66.42	73.75	77.96
mobilenet_v1		224*224*3	604.59	3.41	441.11	4.17
resnet_v1_50		224*224*3	106.3	28.75	173.4	28.68
resnet_v1_101		224*224*3	59.44	45.3	120.38	46.75
resnet_v1_152	uint8	224*224*3	40.4	62.03	91.92	64.61
resnet_v2_50		299*299*3	54.2	55.88	94.6	54.18
resnet_v2_50		224*224*3	102.43	30.36	151.54	32.1
resnet_v2_101		224*224*3	54.37	47.71	103.42	50.16
resnet_v2_152		224*224*3	39.93	66.02	76.18	68.82
vgg_16		224*224*3	19.36	123.58	39.21	160.14
retinaface		640*384*3	332.67	4.4	132.29	4.02
SSD-mobilnet		300*300*3	172.06	13.5	198.18	17.63
mobilenetv2		224*224*3	727.8	3.33	427.35	3.48
yolov2		416*416*3	20.93	66.52	66.97	78.16

NN Accuracy

FrameWork	Mode_Name	Input_size	ADLA Accuracy Status(5000 imagenet dataset)								Board Accuracy on C3(5000 imagenet dataset)			
			Original Model		Int8 Perchannel		Int8 Perlayer		Int16		Int8 Perlayer		Drop	
			Top1	Top5	Top1	Top5	Top1	Top5	Top1	Top5	Top1	Top5	Top1	Top5
PYTORCH	resnet_18	[1,3,224,224]	63.02%	84.78%	62.90%	84.64%	63.14%	84.42%	63.02%	84.72%	62.58%	84.58%	0.32%	0.06%
	resnet_34	[1,3,224,224]	67.70%	88.24%	67.94%	88.24%	67.70%	88.12%	68.14%	88.72%	68.08%	88.36%	-0.14%	-0.12%
	resnet_50	[1,3,224,224]	71.36%	90.12%	71.50%	90.20%	71.34%	90.08%	71.44%	90.22%	71.32%	90.18%	0.18%	0.02%
	resnet_v1_101	[1,3,224,224]	72.84%	91.00%	69.26%	88.98%	68.50%	88.42%	72.76%	91.12%	69.2%	89.38%	0.06%	-0.4%
	squeezenet 1.0	[1,3,224,224]	43.14%	69.16%	43.16%	68.78%	41.68%	67.58%	44.78%	69.74%	43.3%	68.5%	0.14%	0.28%
	SqueezeNet 1.1	[1,3,224,224]	44.38%	69.70%	43.06%	68.32%	41.82%	67.94%	45.14%	69.88%	43.1%	68.5%	-0.04%	-0.18%
	resnext50_32x4d	[1, 3, 300, 300]	72.54%	91.14%	72.86%	91.18%	74.24%	91.16%	73.78%	91.64%	73.45%	91.25%	-0.59%	-0.07%
ONNX	resnet18-v1-7	[1, 3, 224, 224]	66.14%	87.92%	66.26%	87.80%	64.36%	87.12%	66.40%	87.80%	66.48%	87.72%	-0.22%	0.08%
	resnet18-v2-7	[1, 3, 224, 224]	66.70%	87.96%	62.16%	84.26%	64.80%	85.86%	66.68%	87.78%	62.5%	84.54%	0.34%	-0.28%
	resnet34-v1-7	[1, 3, 224, 224]	71.48%	90.16%	71.18%	90.22%	70.56%	89.54%	71.54%	90.34%	71.28%	90.16%	-0.10%	0.06%
	resnet34-v2-7	[1, 3, 224, 224]	71.46%	90.66%	71.58%	90.20%	70.00%	89.56%	71.66%	90.52%	71.48%	90.14%	0.10%	0.06%
	resnet50-v1-7	[1, 3, 224, 224]	72.32%	90.88%	71.60%	90.66%	70.56%	90.16%	72.20%	90.80%	71.18%	90.52%	0.42%	0.14%
	resnet50-v2-7	[1, 3, 224, 224]	73.74%	91.08%	70.96%	89.98%	70.20%	88.64%	73.58%	91.00%	70.90%	89.88%	0.06%	0.10%
	squeezenet1.1-7	[1, 3, 224, 224]	52.00%	75.96%	49.10%	73.70%	48.10%	72.98%	51.40%	75.58%	49.68%	73.52%	-0.58%	0.18%
	vgg16-7	[1, 3, 224, 224]	69.30%	89.48%	71.58%	90.08%	69.34%	89.26%	70.50%	89.96%	70.36%	90.00%	1.22%	0.08%
	vgg19-7	[1, 3, 224, 224]	71.10%	90.74%	71.78%	91.04%	70.92%	90.32%	71.86%	91.00%	71.78%	91.04%	0.00%	0.00%
	vgg19-bn-7	[1, 3, 224, 224]	70.58%	90.08%	71.46%	90.82%	69.62%	89.88%	71.72%	90.86%	71.80%	90.82%	-0.34%	0.00%
CAFFE	inception-v3	[1,3,299,299]	75.94%	92.90%	76.08%	92.68%	75.64%	92.54%	76.02%	93.02%	75.88%	93.13%	0.21%	-0.45%
	inception-v4	[1,3,299,299]	77.28%	93.58%	72.56%	90.18%	72.80%	90.60%	76.88%	93.44%	72.28%	90.83%	0.29%	-0.64%

Amlogic NN Release Source

Document Title	Note
Amlogic_NN_Introduction	Overall introduction
Amlogic_NN_Model_Design_guide	Model design guide
Neural_Network_layer_and_Operation_Support_list	Supported layers and operation (Todo)
Model Conversion User Guide (1.4)	Model conversion guide
adla-toolkit-binary-x.xx.xx.tgz	Model Conversion tool(Contains sdk development kit)
ADLA_SDK API (1.0.0)_CN	Amlogic sdk interface document
amlogic_adla_tool_docker_release.tar.gz	The docker file that has been configured to compile and convert the environment

THANK YOU!